

ECE 461 - HW08

1)

$$y[m] = x[m] + 2x[m-1] + w[m]$$

$$\hookrightarrow w[m] \sim \mathcal{N}(0, \sigma^2)$$

$$\hookrightarrow x[m] = 0 \text{ for } m \neq 1, 2, \quad x[1, 2] \text{ iid. with mean } 0, \text{ variance } E$$

$$\hookrightarrow \text{SNR} = E/\sigma^2$$

1. [Linear Equalizers]
Consider the 2-tap ISI channel with
$$y[n] = x[n] + 2x[n-1] + w[n]$$

where the $w[n]$ are i.i.d. $\mathcal{N}(0, \sigma^2)$ and we define $\text{SNR} = \frac{E}{\sigma^2}$. Only two symbols $x[1]$ and $x[2]$ are sent on this channel, so assume that $x[m] = 0$ for $m \neq 1, 2$. Further, assume that $x[1]$ and $x[2]$ are i.i.d. with mean 0 and variance E . The goal is to detect $x[1]$ from $y[1]$ and $y[2]$, using a linear equalizer based on the sum
$$\hat{y}[1] = c_1 y[1] + c_2 y[2]$$

Find the coefficients c_1 and c_2 of the MMSE equalizer, and the corresponding effective SNR.

$$y[1] = x[1] + w[1] \approx x[1] + z[1]$$

$$y[2] = x[2] + 2x[1] + w[2] = 2x[1] + z[2]$$

$$\text{where } \begin{cases} z[1] = w[1] \\ z[2] = x[2] + w[2] \end{cases}$$

We estimate $x[1]$ using $\hat{y}[1] = c_1 y[1] + c_2 y[2]$

We need to rescale the noise first so $y[1]$ & $y[2]$ have equal variances.

$$\sigma_{z[1]}^2 = \text{Var}\{w[1]\} = \sigma^2$$

$$\sigma_{z[2]}^2 = \text{Var}\{x[2] + w[2]\} = \text{Var}\{x[2]\} + \text{Var}\{w[2]\} = E + \sigma^2$$

$\hookrightarrow$ since x and w are uncorrelated

$$\left. \begin{aligned} \tilde{y}[1] &= y[1] \\ \tilde{y}[2] &= c y[2] \end{aligned} \right\} \quad c^2 \sigma_{z[2]}^2 = \sigma_{z[1]}^2 \Rightarrow c = \sqrt{\frac{\sigma^2}{E + \sigma^2}} = \sqrt{\frac{1}{\text{SNR} + 1}}$$

$\rightarrow$ Now do a matched filter

$$\hat{y}[1] = h[0] \tilde{y}[1] + c h[1] \tilde{y}[2] = \tilde{y}[1] + 2c \tilde{y}[2] = y[1] + 2c^2 y[2]$$

MMSE filter is $\hat{y}[1] = c_1 y[1] + c_2 y[2]$

with $c_1 = 1$, $c_2 = \frac{2}{\text{SNR} + 1}$

$$\hat{y}[1] = x[1] + w[1] + 2c^2 x[2] + 2c^2 2x[1] + 2c^2 w[2]$$

$$= \underbrace{(4c^2 + 1) x[1]}_{\text{signal}} + \underbrace{2c^2 x[2]}_{\text{ISI}} + \underbrace{w[1] + 2c^2 w[2]}_{\text{noise}}$$

$$\text{SNR} = \frac{(4c^2 + 1)^2 E}{4c^4 E + \sigma^2(1 + 4c^4)}$$

$$= \frac{\left(4 \frac{1}{\text{SNR} + 1} + 1\right)^2 \text{SNR}}{4 \left(\frac{1}{\text{SNR} + 1}\right)^2 \text{SNR} + 1 + 4 \left(\frac{1}{\text{SNR} + 1}\right)^2}$$

$$= \frac{(\text{SNR} + 5)^2 \text{SNR}}{4 \text{SNR} + (\text{SNR} + 1)^2 + 4} = \frac{\text{SNR}^2 + 10\text{SNR} + 25}{\text{SNR}^2 + 6\text{SNR} + 5} \text{SNR} = \boxed{\frac{\text{SNR} + 5}{\text{SNR} + 1} \cdot \text{SNR}}$$

2.a) Similar setup as the previous question

$$y[m] = x[m] + h_2 x[m-2] + w[m]$$

$$\left. \begin{aligned} y[1] &= x[1] + w[1] = x[1] + z[1] \\ y[2] &= x[2] + w[2] = z[2] \\ y[3] &= h_2 x[1] + x[3] + w[3] = h_2 x[1] + z[3] \end{aligned} \right\} \begin{aligned} z[1] &= w[1] \\ z[2] &= x[2] + w[2] \\ z[3] &= x[3] + w[3] \end{aligned}$$

2.b)

$$\sigma_{z[1]}^2 = \sigma^2, \quad \sigma_{z[2]}^2 = E + \sigma^2, \quad \sigma_{z[3]}^2 = E + \sigma^2$$

↳ To match the variances, rescale $y[2]$ and $y[3]$ with

$$\left. \begin{aligned} \tilde{y}[1] &= y[1] \\ \tilde{y}[2] &= c y[2] \\ \tilde{y}[3] &= c y[3] \end{aligned} \right\} c^2 \sigma_{z[2]}^2 = c^2 \sigma_{z[3]}^2 = \sigma_{z[1]}^2 \Rightarrow c^2 = \frac{1}{1 + \text{SNR}}$$

$$\tilde{y}[1] = \underbrace{1}_{\tilde{h}[1]} \cdot x[1] + \underbrace{0}_{\tilde{h}[2]} \cdot x[1] + \tilde{z}[1], \quad \tilde{y}[2] = \underbrace{0}_{\tilde{h}[2]} \cdot x[1] + \tilde{z}[2], \quad \tilde{y}[3] = \underbrace{c h_2}_{\tilde{h}[3]} x[1] + \tilde{z}[3]$$

Doing the matched filtering, we obtain the MMSE filter as

$$\hat{y}_{\text{MMSE}}[1] = 1 \cdot y[1] + 0 \cdot (c y[2]) + c h_2 \cdot (c y[3]) = y[1] + c^2 h_2 y[3]$$

$$c_1 = 1, \quad c_2 = 0, \quad c_3 = c^2 h_2 = \frac{h_2}{\text{SNR} + 1}$$

2.c)

$$\hat{y}_{\text{MMSE}}[1] = \underbrace{(1 + c^2 h_2)}_{\text{signal}} x[1] + \underbrace{z[1] + c^2 h_2 z[3]}_{\text{ISI + noise}}$$

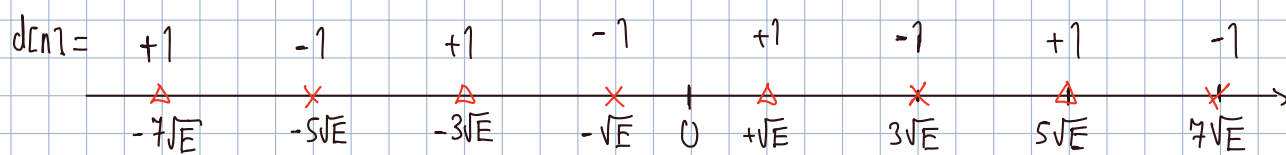
$$\text{SNR} = \frac{\left(1 + \frac{h_2^2}{\text{SNR}+1}\right)^2 E}{\sigma^2 + \left(\frac{h_2}{\text{SNR}+1}\right)^2 (E + \sigma^2)} = \frac{(\text{SNR}+1+h_2^2)^2 \text{SNR}}{(\text{SNR}+1)^2 + h_2^2 (\text{SNR}+1)}$$

$$= \text{SNR} \frac{(\text{SNR}+1+h_2^2)^2}{(\text{SNR}+1)(\text{SNR}+1+h_2^2)} = \boxed{\left[1 + \frac{h_2^2}{\text{SNR}+1}\right] \cdot \text{SNR}}$$

$$3) \quad y[n] = x[n] + \frac{g}{10} x[n-1] + w[n]$$

↳ with $E[w[n]] = 0$, $\text{Var}[w[n]] = \sigma^2$

↳ Transmission done using the Tomlinson-Harashima precoding with $d[n] = \pm 1$



→ We extend the constellation to avoid excessive energy transmission.

↳ For $d[n] = +1$: $v \in \{-7, -3, 1, 5, 9\} \sqrt{E}$

↳ For $d[n] = -1$: $v \in \{-5, -1, 3, 7\} \sqrt{E}$

* Transmitter chooses $v[n]$ to minimize the magnitude of the transmitted symbol:

$$|x[n]| = |v[n] - I[n]| \leq 2\sqrt{E}$$

* For $y[n] = x[n] + 0.9x[n-1] + w[n]$, interference term is $I[n] = 0.9x[n-1]$

Given data sequence $d = [1, -1, 1, 1]$, we calculate the transmission iteratively

$$n=1 : d[1] = 1, I[1] = 0$$

$$x[1] = 1 - 0 = 1$$

↳ closest valid $v[1]$ to $I[1]$ is 1

$$n=2 : d[2] = -1, I[2] = 0.9x[1] = 0.9$$

↳ Valid $v[2]$ for $d=-1$ are $\dots, -5, -1, 3, 7, \dots$

↳ To keep $x[2]$ within $[-2, 2]$ we choose $v[2] = -1$

$$x[2] = -1 - 0.9 = -1.9$$

$$n=3: d[3] = +1, I[3] = 0.9 \times (-1.9) = -1.71$$

↳ Valid $V[3]$ for $d = +1$ are $\dots -3, 1, 5 \dots$

↳ We choose $V[3] = -3$ for

$$X[3] = -3 - (-1.71) = -1.29 \text{ to stay within } [-2, 2].$$

$$n=4: d[4] = +1, I[4] = 0.9(-1.29) = -1.161$$

↳ Valid $V[3]$ for $d = +1$ are $\dots -3, 1, 5 \dots$

↳ We choose $V[3] = -3$ for

$$X[3] = -3 - (-1.161) = -1.839 \text{ to stay within } [-2, 2].$$

Sequence of precoded symbols are $\mathbf{X} = [1, -1.9, -1.29, -1.161]$

3.b) Since TH precoding turns the system into an ISI free AWGN channel, we can use ML to pick the closest point in the extended constellation.

$$Y[1] = 0.8628 \rightarrow \text{Nearest odd integer is } 1 \rightarrow d[1] = 1$$

$$Y[2] = 0.4499 \rightarrow \text{Nearest odd integer is } 1 \rightarrow d[2] = 1$$

$$Y[3] = 0.6842 \rightarrow \text{Nearest odd integer is } 1 \rightarrow d[3] = 1$$

$$Y[4] = 2.8959 \rightarrow \text{Nearest odd integer is } 3 \rightarrow d[4] = -1$$

$$Y[5] = 3.0817 \rightarrow \text{Nearest odd integer is } 3 \rightarrow d[5] = -1$$

$$Y[6] = 2.7890 \rightarrow \text{Nearest odd integer is } 3 \rightarrow d[6] = -1$$

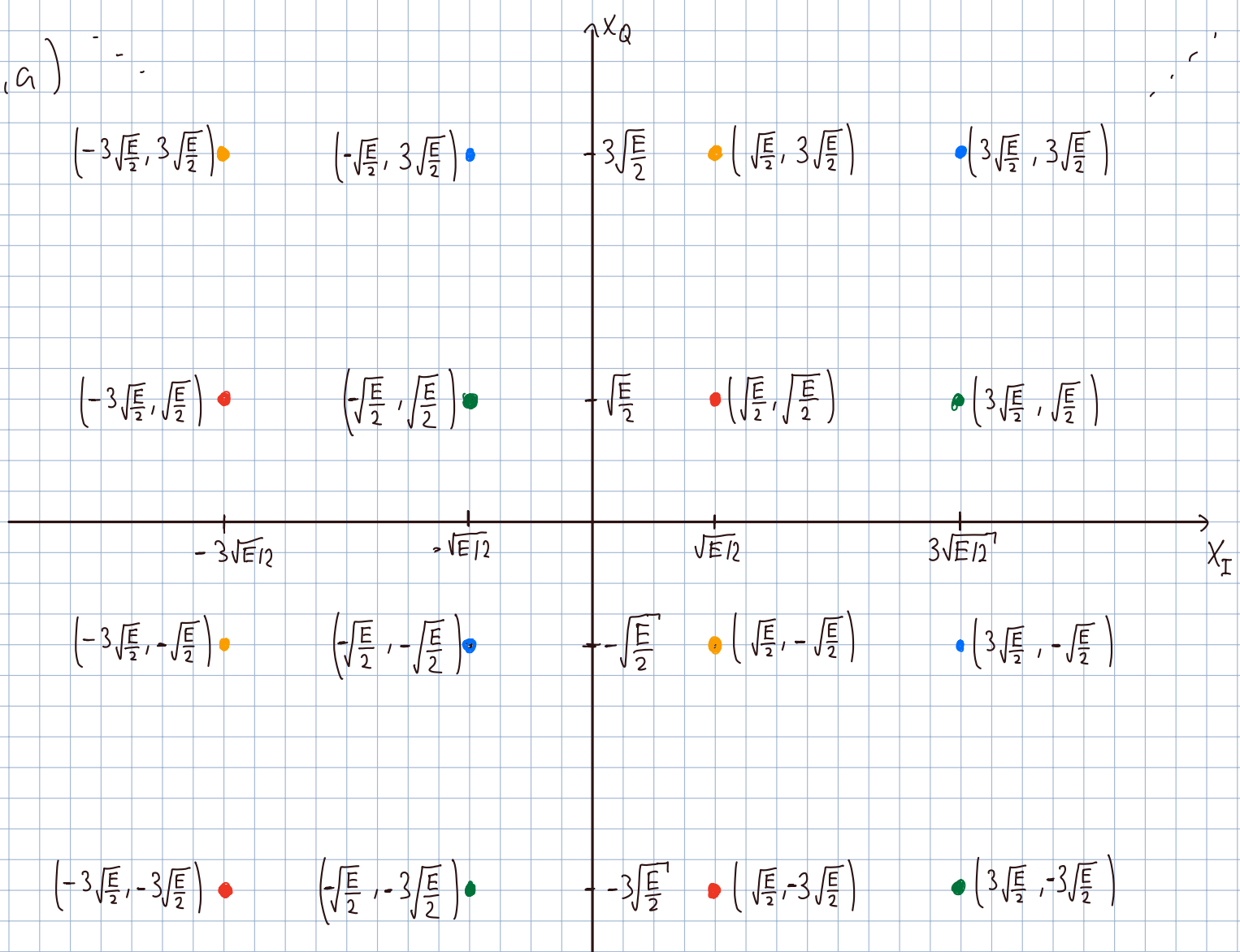
$$Y[7] = 2.6964 \rightarrow \text{Nearest odd integer is } 3 \rightarrow d[7] = -1$$

$$Y[8] = 0.7235 \rightarrow \text{Nearest odd integer is } 1 \rightarrow d[8] = 1$$

$$Y[9] = -0.8008 \rightarrow \text{Nearest odd integer is } -1 \rightarrow d[9] = -1$$

Estimated data sequence is $\mathbf{d} = [1, 1, 1, -1, -1, -1, -1, 1, -1]$

4, a)



* We basically extend the constellation by shifting the central pattern by $4\sqrt{\frac{E}{2}}$ in every direction.

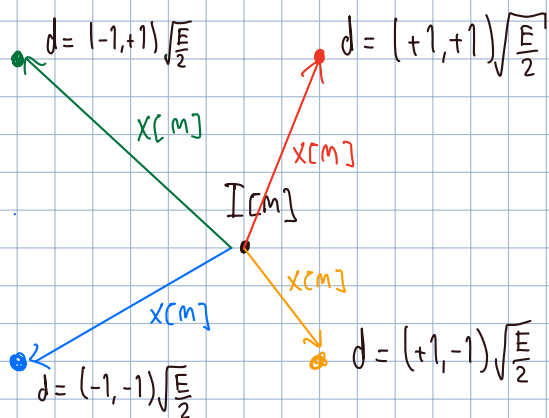
● $(1+4m, 1+4k) \sqrt{\frac{E}{2}}, m, k \in \mathbb{Z}$

● $(-1+4m, 1+4k) \sqrt{\frac{E}{2}}, m, k \in \mathbb{Z}$

● $(-1+4m, -1+4k) \sqrt{\frac{E}{2}}, m, k \in \mathbb{Z}$

● $(1+4m, -1+4k) \sqrt{\frac{E}{2}}, m, k \in \mathbb{Z}$

4.b) In 4-QPSK scheme, each $I[m]$ will be inside one of the 4 $x[m]$ in a square of sidelength $2\sqrt{\frac{E}{2}}$



→ Diagram on the left, shows possible $x[m]$ based on data sequence.

→ Since $\|x_I\|_2 \leq 2\sqrt{\frac{E}{2}}$, $\|x_Q\| \leq 2\sqrt{\frac{E}{2}}$ where x_I, x_Q are the components of x (either in-phase & quadrature or $x[1]$ & $x[2]$):

$$E_{\max} = x_I^2 + x_Q^2 = \left(2\sqrt{\frac{E}{2}}\right)^2 + \left(2\sqrt{\frac{E}{2}}\right)^2 = \boxed{4E}$$